

Olympiad Test Paper — Science (Class 7)

Chapter 19: Earth, Moon and the Sun

Subject	Science (NCERT Class 7)
Chapter	19 — Earth, Moon and the Sun
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. No calculator is needed — all numbers are designed to work by hand. Many questions need two to four reasoning steps; watch for traps that mix up rotation with revolution, phases with eclipses, or distance with tilt. Do not write on this paper — mark answers on your answer sheet.

1. Day and night on Earth are caused by:

- (A) Earth's revolution around the Sun (B) the tilt of Earth's axis
(C) Earth's rotation on its axis (D) the Moon blocking sunlight

2. A friend says, "It is night in India, so the Sun has switched off for a while." The correct reply is:

- (A) the Sun keeps shining; India has simply rotated to the side facing away from the Sun
(B) the Sun does switch off at night and relights at dawn
(C) the Moon hides the Sun every night
(D) the Sun has moved to the other side of Earth

3. Earth completes one full spin on its axis in about:

- (A) 365 days (B) 27.3 days
(C) 1 hour (D) 24 hours

4. Which single motion is responsible for the year, and which for the day?

- (A) rotation gives the year; revolution gives the day
(B) revolution gives the year; rotation gives the day
(C) both the year and the day come from rotation
(D) both the year and the day come from revolution

5. Earth's axis is tilted at about:

- (A) 23.5 degrees (B) 90 degrees
(C) 0 degrees (D) 45 degrees

6. Earth is best described in shape as:

- (A) a perfect sphere (B) a flat disc
(C) a slightly flattened sphere (squished at the poles)
(D) a cube with rounded edges

7. As Earth orbits the Sun through the year, the direction in which its axis points:

- (A) flips upside-down every six months (B) stays fixed, always leaning the same way
(C) spins to follow the Sun (D) becomes perfectly upright in summer

8. The main reason India has summer in June is that in June the Northern Hemisphere:

- (A) is closest to the Sun (B) is tilted away from the Sun
(C) spins faster than in winter
(D) is tilted toward the Sun, receiving more direct rays

9. Earth is actually a little **closer** to the Sun in January than in July. This tells us that seasons are caused mainly by:

- (A) Earth's distance from the Sun (B) the Moon's gravity
(C) the tilt of Earth's axis, not its distance (D) the speed of Earth's rotation

10. When the Northern Hemisphere has summer, the Southern Hemisphere (e.g. Australia) has:

- (A) summer at the same time (B) winter
(C) no season at all (D) two summers

11. Slanting sunlight in winter feels less warm than the overhead sunlight of summer because slanting rays:

- (A) spread the same energy over a larger area, so each spot gets less heat
(B) are a different colour (C) travel a shorter distance
(D) carry more energy per area

12. If Earth's axis were straight up (no tilt at all), the result would be:

- (A) no day and night (B) summer everywhere, always
(C) a much longer year
(D) no seasons — each place would have nearly the same weather all year

13. Earth takes 365 days **and about 6 extra hours** to orbit the Sun. To keep the calendar in step with the seasons, we:

- (A) add an extra day (February 29) every four years
(B) drop one day every year (C) add an extra month every year
(D) ignore the extra hours forever

14. A leap year is added every four years because the leftover quarter-days add up to:

- (A) one full week after four years (B) one full day after four years
(C) one full hour after four years (D) one full month after four years

15. Earth is divided into 24 time zones because it rotates:

- (A) 24 degrees in 360 hours (B) once every year
(C) 360 degrees in 24 hours, so each 15-degree slice differs by one hour
(D) backwards near the poles

16. India is geographically wide, yet the whole country keeps a single clock time (IST). This is mainly for:

- (A) astronomical accuracy at every village (B) matching the Moon's phases
(C) following the Southern Hemisphere
(D) convenience, so the whole nation shares one standard time

17. To an observer in India, the Sun appears to rise roughly in the:

- (A) west and set in the east (B) east and set in the west
(C) north and set in the south (D) same spot all day

18. The Sun seems to travel across the sky from morning to evening. The real reason is that:

- (A) Earth rotates, carrying us past the steady Sun
(B) the Sun orbits Earth once a day (C) the Sun is pushed by the Moon
(D) Earth revolves once a day

19. Winds and cyclones curve as they move because Earth is rotating. In the Northern Hemisphere, moving air is deflected to the:

- (A) left (B) straight up
(C) right (D) it is never deflected

20. A cyclone spins counter-clockwise over India but clockwise over Australia. The best explanation is:

- (A) Earth's rotation deflects winds in opposite directions in the two hemispheres
(B) the two storms obey different laws of physics
(C) Australia is closer to the Moon
(D) only Indian cyclones are affected by the Sun

21. The Pole Star (Dhruv Tara) appears almost stationary all night because it lies nearly in line with Earth's:

- (A) equator (B) orbit around the Sun
(C) magnetic field (D) rotational axis

22. To find the Pole Star, you first locate the seven bright stars of the Saptarishi and then:

- (A) look straight down at the horizon
(B) extend the line of the two pointer stars at the bowl's edge

(C) wait for it to twinkle red

(D) follow the Moon's path

23. Once you have found the Pole Star and are facing it, the direction on your right hand is:

(A) west

(B) south

(C) east

(D) north

24. The Moon is best described as:

(A) a star that orbits Earth

(B) Earth's only natural satellite

(C) a planet of the Sun

(D) a piece of the Sun

25. We can see the Moon at night because it:

(A) reflects sunlight that falls on it

(B) makes its own light like a star

(C) glows with stored heat

(D) is lit by Earth's light only

26. A student claims the Moon is a tiny star because it shines. The flaw in this claim is that:

(A) stars are smaller than the Moon

(B) the Moon is hotter than a star

(C) stars never appear at night

(D) the Moon does not produce its own light; it only reflects the Sun's light

27. There is no wind, no weather, and no sound on the Moon mainly because the Moon has:

(A) too much water

(B) a very fast spin

(C) no atmosphere (no air)

(D) a strong magnetic field

28. A person who weighs 60 kg-force on Earth would weigh much less on the Moon because the Moon's gravity is about:

(A) the same as Earth's

(B) one-sixth of Earth's

(C) six times Earth's

(D) zero

29. The Moon's surface is covered with bowl-shaped pits called craters, which were formed mainly by:

(A) volcanoes erupting water

(B) ocean waves

(C) impacts of asteroids and meteoroids over billions of years

(D) wind erosion

30. The changing shapes of the Moon through a month (new, crescent, quarter, gibbous, full) are called its:

(A) phases

(B) eclipses

(C) tides

(D) craters

31. The Moon's phases happen because, as the Moon orbits Earth, we see:

(A) Earth's shadow sweep across it each night

(B) the Moon actually change its shape

(C) clouds cover different parts of it

(D) different amounts of its sunlit half

- 32.** The full cycle of the Moon's phases (new Moon back to new Moon) takes about:
- (A) 24 hours
 - (B) 29.5 days
 - (C) 7 days
 - (D) 365 days
- 33.** The Moon is roughly between the Sun and a viewer on Earth, with its lit side facing away from us. We therefore see a:
- (A) new Moon (nearly invisible)
 - (B) full Moon
 - (C) first-quarter Moon
 - (D) lunar eclipse
- 34.** A common mistake is to say "the Moon's phases are caused by Earth's shadow falling on the Moon." This is wrong because Earth's shadow on the Moon actually causes a:
- (A) solar eclipse
 - (B) new Moon every night
 - (C) lunar eclipse
 - (D) spring tide
- 35.** We always see the same face of the Moon from Earth because the Moon:
- (A) does not rotate at all
 - (B) rotates once on its axis in the same time it takes to orbit Earth
 - (C) is held still by Earth's magnetism
 - (D) spins much faster than it orbits
- 36.** The so-called "dark side" of the Moon is better called the "far side" because it:
- (A) never receives any sunlight
 - (B) is always in Earth's shadow
 - (C) is colder than the rest of the Moon by law
 - (D) actually receives sunlight too, but we simply never see it from Earth
- 37.** During a solar eclipse, the body that comes between the Sun and Earth is the:
- (A) Moon
 - (B) Mars
 - (C) a comet
 - (D) Jupiter
- 38.** The Moon is about 400 times smaller than the Sun, yet it can just cover the Sun during a solar eclipse. This is possible because the Moon is:
- (A) brighter than the Sun
 - (B) about 400 times closer to us than the Sun, so they look the same size
 - (C) made of denser material
 - (D) larger than the Sun in reality
- 39.** A lunar eclipse occurs when:
- (A) the Moon passes between the Sun and Earth
 - (B) clouds cover the Moon
 - (C) Earth comes between the Sun and the Moon and casts its shadow on the Moon
 - (D) the Sun passes between Earth and the Moon
- 40.** During a total lunar eclipse the Moon often glows a coppery red instead of going fully black. This is because Earth's atmosphere:
- (A) makes its own red light
 - (B) sets the Moon on fire
 - (C) reflects red light from the oceans

(D) bends red sunlight into Earth's shadow, falling on the Moon

41. A solar eclipse can only happen at the new-Moon position and a lunar eclipse only at the full-Moon position because eclipses need the Sun, Moon and Earth to be:

- (A) at right angles
- (B) very nearly in a straight line
- (C) at their farthest apart
- (D) spinning fast

42. The daily rise and fall of sea level along coasts — the tides — are caused mainly by the gravitational pull of the:

- (A) Moon (helped by the Sun)
- (B) Sun only
- (C) other planets
- (D) Earth's rotation alone

43. Most coastal places get about two high tides and two low tides each day. One reason there are **two** high-tide bulges is that:

- (A) the Sun pulls twice a day
- (B) the Moon orbits twice a day
- (C) the tides come only from rivers
- (D) there is a bulge facing the Moon and another on the opposite side of Earth

44. Extra-high "spring tides" occur at new Moon and full Moon because at those times the Sun and Moon:

- (A) pull at right angles, partly cancelling
- (B) stop pulling on the oceans
- (C) line up so their pulls add together
- (D) move to opposite sides of the galaxy

45. The weaker "neap tides" occur at the quarter Moons because then the Sun and Moon pull on the oceans:

- (A) at right angles to each other, so the pulls partly cancel
- (B) from the same direction
- (C) not at all
- (D) twice as hard

46. The first humans to walk on the Moon, in 1969, arrived aboard the mission named:

- (A) Chandrayaan-3
- (B) Apollo 11
- (C) Mangalyaan
- (D) Sputnik 1

47. A star such as our Sun produces its enormous light and heat by:

- (A) burning wood and coal
- (B) reflecting light from other stars
- (C) nuclear fusion of hydrogen into helium in its core
- (D) a chemical reaction with oxygen

48. The statement that best fits modern astronomy is:

- (A) the Sun is a uniquely special, one-of-a-kind object
- (B) the Sun is a planet that gives out light
- (C) the Sun is the largest star in the universe
- (D) the Sun is an ordinary, fairly average star — one of billions

49. The star nearest to Earth is:

- (A) the Pole Star
- (C) Sirius

- (B) the Sun
- (D) Alpha Centauri

50. Stars appear to twinkle while planets shine with a steadier light. The reason is that:

- (A) starlight, coming from a near-point, is bent unevenly by Earth's wobbly atmosphere, while a planet's wider disc averages the wobble out
- (B) stars are alive and planets are not
- (C) planets make their own steady light
- (D) stars are closer than planets

51. A recognisable pattern of stars in the night sky, such as the hunter Kalpurush (Orion), is called a:

- (A) galaxy
- (B) comet
- (C) constellation
- (D) solar system

52. The stars that make up a single constellation are usually:

- (A) all the same distance from Earth
- (B) at very different distances, only appearing to line up from Earth
- (C) physically close together in space
- (D) all part of our Solar System

53. The number of planets in the Solar System is:

- (A) 7
- (B) 9
- (C) 10
- (D) 8

54. At the centre of the Solar System, holding all the planets in their orbits with its gravity, is the:

- (A) Earth
- (B) Moon
- (C) Sun
- (D) Pole Star

55. The four inner planets of the Solar System are:

- (A) Mercury, Venus, Earth, Mars
- (B) Jupiter, Saturn, Uranus, Neptune
- (C) Sun, Earth, Moon, Mars
- (D) Pluto, Ceres, Eris, Sedna

56. Compared with the inner planets, the outer planets (Jupiter, Saturn, Uranus, Neptune) are:

- (A) small and rocky with solid surfaces
- (B) huge gas or ice giants with no solid surface to stand on
- (C) closer to the Sun
- (D) all without any moons

57. Pluto is no longer counted among the planets; it is now classed as a:

- (A) star
- (B) comet
- (C) dwarf planet
- (D) moon of Neptune

58. Rocky chunks that mostly orbit the Sun in a belt between Mars and Jupiter are called:

- (A) asteroids
- (B) comets
- (C) meteors
- (D) galaxies

59. A "shooting star" is really a meteor — a small piece of space rock that:

- (A) is a distant star falling from the sky
- (B) orbits the Sun like a planet
- (C) is a piece of the Moon
- (D) burns up brightly as it speeds through Earth's atmosphere

60. A light-year, used to state the huge distances between stars, is a unit of:

- (A) time
- (B) distance
- (C) mass
- (D) temperature

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 19: Earth, Moon and the Sun

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	D	31	D	41	B	51	C
2	A	12	D	22	B	32	B	42	A	52	B
3	D	13	A	23	C	33	A	43	D	53	D
4	B	14	B	24	B	34	C	44	C	54	C
5	A	15	C	25	A	35	B	45	A	55	A
6	C	16	D	26	D	36	D	46	B	56	B
7	B	17	B	27	C	37	A	47	C	57	C
8	D	18	A	28	B	38	B	48	D	58	A
9	C	19	C	29	C	39	C	49	B	59	D
10	B	20	A	30	A	40	D	50	A	60	B

Answer-key distribution: A = 15, B = 17, C = 15, D = 13.

Worked Solutions

- (C)** Day and night come from Earth's **rotation** — its spin on its own axis once every 24 hours. The side facing the Sun has day; the opposite side has night. Revolution (a year long) gives seasons, not day/night; the Moon blocks the Sun only during a rare solar eclipse.
- (A)** The Sun shines steadily all the time. Night happens because the part of Earth where India sits has rotated to face away from the Sun — the Sun does not switch off, and the Moon is not hiding it.
- (D)** One full spin (rotation) of Earth on its axis takes about 24 hours — that is exactly why a day is 24 hours long. (365 days is one revolution; 27.3 days is the Moon's orbit.)
- (B)** Revolution — going once around the Sun (≈ 365.25 days) — gives the year. Rotation — one spin on the axis (24 h) — gives the day. Mixing these two up is the classic trap.
- (A)** Earth's axis is tilted about 23.5° away from the upright (perpendicular to its orbit). This fixed tilt is what produces the seasons.

- 6. (C)** Earth is a slightly flattened sphere — an oblate spheroid, a tiny bit squished at the poles and bulging at the equator. It is neither a perfect sphere nor flat.
- 7. (B)** Earth's axis keeps pointing in the same fixed direction throughout the year (roughly toward the Pole Star). Because the lean stays fixed while Earth orbits, different hemispheres face the Sun at different times — that is what makes seasons.
- 8. (D)** In June the Northern Hemisphere is tilted **toward** the Sun, so India receives more direct (overhead) rays and longer days → summer. It is not about being closest to the Sun.
- 9. (C)** Earth is actually closest to the Sun in January (Indian winter) and farthest in July (Indian summer). Since the seasons run "backwards" to the distance, distance cannot be the cause — the 23.5° **tilt** is.
- 10. (B)** When the Northern Hemisphere tilts toward the Sun (its summer), the Southern Hemisphere tilts away → winter. So Australia has winter when India has summer. The two hemispheres always have opposite seasons.
- 11. (A)** Slanting rays strike the ground at an angle, so the same beam of energy is spread over a larger patch of ground — each square metre receives less heat. Overhead (direct) rays concentrate the energy → warmer. (Slanting rays carry less, not more, energy per area.)
- 12. (D)** Seasons come entirely from the 23.5° tilt. With no tilt, no hemisphere would ever lean toward or away from the Sun, so each place would get nearly the same sunlight all year — no summer-winter swing.
- 13. (A)** The leftover ≈ 6 hours each year add up to one whole day every four years. We absorb it by adding February 29 in a leap year, keeping the calendar lined up with the seasons.
- 14. (B)** Four years $\times$ 6 extra hours = 24 hours = one full day. That accumulated day is added as February 29.
- 15. (C)** Earth turns a full 360° in 24 hours, i.e. 15° per hour. Dividing the globe into 24 slices of 15° each gives 24 time zones, neighbouring zones differing by one hour.
- 16. (D)** A wide country such as India spans enough longitude that the Sun is genuinely "ahead" in the east and "behind" in the west, but running every town on its own clock would be chaos. So the whole nation keeps one standard time (IST) purely for convenience.
- 17. (B)** Because Earth rotates from west to east, the Sun appears to come up in the **east** and go down in the **west**. (It only *appears* to move; really we are turning.)

18. (A) The Sun stays essentially fixed; Earth's rotation carries us along, so the Sun seems to sweep from east to west across the sky during the day. This is rotation (a day), not revolution (a year).

19. (C) Earth's rotation deflects moving air — the Coriolis effect. In the Northern Hemisphere the deflection is to the **right**. (It is to the left in the Southern Hemisphere.)

20. (A) Same physics, opposite directions: the Coriolis deflection is rightward in the Northern Hemisphere and leftward in the Southern, so cyclones spin counter-clockwise over India but clockwise over Australia. They do not obey different laws.

21. (D) Earth's spin axis points almost exactly at the Pole Star, so as Earth turns, that star appears fixed while all the others wheel around it. It is the **rotational axis** (geographic, not magnetic) that lines up with it.

22. (B) Locate the Saptarishi (Big Dipper/Ursa Major); the two stars at the outer edge of its "bowl" are the pointer stars. Extend the line through them about five times that gap and you reach the Pole Star.

23. (C) Facing the Pole Star you are facing **north**. North ahead means **east** is on your right, west on your left, and south behind you.

24. (B) The Moon is Earth's only natural satellite — a body held by Earth's gravity that orbits it. It is not a star (it makes no light of its own) and not a planet (planets orbit the Sun directly).

25. (A) The Moon has no light of its own; we see it because it **reflects** sunlight falling on it. (It does not glow from stored heat, nor is it lit only by Earth.)

26. (D) A star produces its own light by nuclear fusion; the Moon does not. The Moon only shines by reflecting the Sun's light, so calling it a "tiny star" is wrong.

27. (C) The Moon has almost no atmosphere — no air. With no air there can be no wind, no weather, and no medium to carry sound. (It also has no liquid water.)

28. (B) The Moon's surface gravity is about one-sixth of Earth's. So a 60 kg-force person would weigh roughly $60 \div 6 = 10$ kg-force there — and could jump much higher. (Mass stays the same; weight is what drops.)

29. (C) The Moon's craters are bowl-shaped pits gouged out by the impacts of asteroids and meteoroids over billions of years. With no air and no water, there is little erosion to wear them away.

30. (A) The Moon's changing monthly shapes — new, crescent, quarter, gibbous, full — are its **phases**. (Eclipses are a different, occasional event; tides are ocean movements; craters are surface pits.)

- 31. (D)** Half the Moon is always sunlit. As the Moon orbits Earth, we view that sunlit half from changing angles, so we see different *amounts* of it lit — that is what makes the phases. The Moon does not really change shape, and Earth's shadow is not involved.
- 32. (B)** One complete cycle of phases — new Moon back to new Moon — takes about 29.5 days. (27.3 days is the Moon's orbit relative to the stars; the phase cycle is a little longer because Earth has also moved around the Sun.)
- 33. (A)** When the Moon lies roughly between us and the Sun, its lit side faces *away* from Earth, so we see essentially no lit part → a **new Moon** (nearly invisible). (At full Moon the Moon is on the far side of Earth from the Sun.)
- 34. (C)** Phases are *not* caused by Earth's shadow. Earth's shadow falling on the Moon is what produces a **lunar eclipse** — a separate, occasional event that happens only at full Moon.
- 35. (B)** The Moon is "tidally locked": it spins once on its axis in exactly the time it takes to orbit Earth (about 27.3 days). Because rotation and orbit are synchronised, the same face always points toward us.
- 36. (D)** The "far side" is wrongly called the "dark side." It actually receives just as much sunlight as the near side over a month — we simply never see it from Earth because of tidal locking. It is not permanently dark or in shadow.
- 37. (A)** In a solar eclipse the **Moon** moves directly between the Sun and Earth, blocking the Sun and casting its shadow on Earth. (Mars and Jupiter are far beyond Earth's orbit and cannot get between us and the Sun; comets effectively never line up this way.)
- 38. (B)** The Moon is about 400 times smaller than the Sun but also about 400 times closer to us. Those two factors cancel, so the Moon and Sun look almost the same size in our sky — which is why the Moon can just cover the Sun.
- 39. (C)** A lunar eclipse occurs when **Earth** lies between the Sun and the Moon, so Earth's shadow falls on the Moon. (The Moon between Sun and Earth would be a *solar* eclipse.)
- 40. (D)** During totality the Moon turns coppery red rather than black because Earth's atmosphere bends (refracts) red sunlight into Earth's shadow, and that reddened light lands on the Moon — the same reddening that makes sunsets red.
- 41. (B)** Both eclipses need the Sun, Moon, and Earth almost exactly in a straight line. That line-up only happens at new Moon (Moon between Sun and Earth → solar eclipse) or full Moon (Earth between Sun and Moon → lunar eclipse).
- 42. (A)** Tides are mainly the work of the **Moon's** gravitational pull on the oceans, with the Sun adding a smaller effect. (Other planets are far too distant to matter; rotation alone

would not raise the water.)

43. (D) There is one water bulge on the side facing the Moon (pulled toward it) and a second bulge on the opposite side (because the Moon pulls Earth's solid body more strongly than the far ocean). As Earth rotates, a coast passes through both bulges → about two high tides a day.

44. (C) At new Moon and full Moon the Sun and Moon line up with Earth, so their gravitational pulls add together, giving extra-high (and extra-low) **spring tides**.

45. (A) At the quarter Moons the Sun and Moon are at right angles as seen from Earth, so their pulls partly cancel. The result is the gentler **neap tides**.

46. (B) The first crewed Moon landing was **Apollo 11** in July 1969, when Neil Armstrong and Buzz Aldrin walked on the Moon. (Chandrayaan-3 and Mangalyaan are later Indian missions; Sputnik 1 was the first artificial satellite.)

47. (C) A star shines by **nuclear fusion** — in its core, hydrogen fuses into helium, releasing enormous light and heat. It is not ordinary burning (no oxygen out there) and it does not merely reflect other stars' light.

48. (D) Astronomers regard the Sun as an ordinary, fairly average star — one of billions in our galaxy. What is special is our distance from it (right for liquid water), not the star itself. It is certainly not a planet, nor the largest star.

49. (B) The **Sun** is by far the nearest star — only about 8 light-minutes away. The Pole Star, Sirius, and Alpha Centauri are all many light-years away.

50. (A) A star is so distant it appears as a point; its thin beam is bent unevenly by Earth's turbulent atmosphere, so it seems to flicker — twinkling. A planet, being nearer, shows a tiny disc, whose many points of light average out the wobble, so it glows steadily. (Planets do not make their own light; nearness does not cause twinkling — it reduces it.)

51. (C) A recognisable pattern of stars in the sky — like Kalpurush (Orion) or the Saptarishi — is a **constellation**. (A galaxy is a vast collection of billions of stars; a comet is an icy body; a solar system is a star with its orbiting bodies.)

52. (B) The stars in a constellation usually lie at very different distances from Earth; they only *appear* to form a pattern because they happen to line up along our line of sight — like near trees and far mountains appearing side by side in a photo.

53. (D) The Solar System has **8** planets, Mercury through Neptune. Pluto used to be called the ninth but was reclassified as a dwarf planet in 2006, leaving eight.

54. (C) The **Sun** sits at the centre of the Solar System; its gravity holds all the planets — including Earth — in their orbits. (Earth and the Moon both orbit; the Pole Star is a distant

star, not part of the Solar System.)

55. (A) The four inner (rocky) planets are **Mercury, Venus, Earth, Mars**. The Jupiter–Neptune group are the outer giants; the Sun is a star and the Moon is a satellite, so the other lists are wrong.

56. (B) The outer planets — Jupiter, Saturn, Uranus, Neptune — are huge gas or ice giants with no solid surface to stand on. They are far from the Sun (not closer) and do have many moons.

57. (C) Pluto was reclassified in 2006 as a **dwarf planet** because it has not cleared its orbital neighbourhood. It is not a star, comet, or a moon of Neptune.

58. (A) **Asteroids** are rocky chunks that mostly orbit the Sun in the asteroid belt between Mars and Jupiter. (Comets are icy bodies with tails; meteors are streaks of burning rock in our atmosphere; a galaxy is a star system far larger than the Solar System.)

59. (D) A "shooting star" is a **meteor** — a small piece of space rock that burns up brightly from friction as it speeds through Earth's atmosphere. It is not a real star falling, and if a piece survives to reach the ground it is then called a meteorite.

60. (B) A light-year is the **distance** light travels in one year (about 9.5 trillion km). Despite the word "year," it measures distance, not time, mass, or temperature.